

Each question is worth 5 points for a total of 120 points. The extra credit can get you an additional 5 points making the highest possible score 125 out of 120. Write answers CLEARLY. If I cannot determine your answer, the problem will receive a score of 0.

Problem 1

Convert each angle in degrees to radians. Express your answer as a multiple of π .

- a) -120°
- b) 180°
- c) 225°
- d) 90°

Problem 2

Let s denote the length of the arc of a circle of radius r subtended by the central angle θ . Find the missing quantity.

- a) $r = 7$ meters, $\theta = 3.1$, $s = ?$
- b) $r = \pi$ inches, $\theta = ?$, $s = \frac{2\pi^2}{3}$ inches
- c) $r = ?$, $\theta = \frac{\pi}{4}$, $s = 2\pi$ feet

Problem 3

Find the exact value.

- a) $\sin \frac{7\pi}{6}$
- b) $\sin \frac{-7\pi}{6}$
- c) $\cot \frac{\pi}{3}$

Problem 4

Find the exact value $\frac{\tan 70^\circ - \tan 10^\circ}{1 + \tan 70^\circ \tan 10^\circ}$.

Problem 5

Find the exact value $\sin[\sin^{-1} \frac{8}{17} - \cos^{-1}(-\frac{4}{5})]$.

Problem 6

Find the exact value $\tan \frac{7\pi}{8}$.

Problem 7

Establish the identity and SHOW ALL YOUR WORK. If I cannot tell how you got from one step to the next, you will LOSE POINTS.

$$\tan(3\theta) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

Problem 8

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\sin^2 \theta = \frac{3}{4}$.

Problem 9

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin^2 \theta = \sin \theta + 1$.

Problem 10

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\cos(2\theta) = \cos \theta$.

Problem 11

To measure the height of Lincoln's caricature on Mt. Rushmore, two sightings 200 feet from the base of the mountain are taken. If the angle of elevation to the bottom of Lincoln's face is 30° and the angle of elevation to the top is 60° , what is the height of Lincoln's face?

Problem 12

Solve the triangle $c = 4$, $b = 8$, $\gamma = 30^\circ$.

Problem 13

Find the area of the triangle $a = 4\sqrt{3}$, $b = 8$, $\gamma = \frac{\pi}{4}$.

Problem 14

Find the area of the triangle $a = 5$, $b = 4$, $c = 3$.

Problem 15

The displacement d (in meters) of an object at time t (in seconds) is given by the equation $d = -4\sin(7t)$. Find the amplitude, period, and frequency.

Problem 16

The point $(8, 15)$ is on the terminal side of an angle θ . Find the exact value of the six trigonometric functions.

Problem 17

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \cos^2 \theta + \cos \theta = 1$.

Problem 18

Let $\sin \theta = \frac{7}{25}$ where θ is in the second quadrant. Find the exact value of the other five trigonometric functions.

Problem 19

Find the exact value of $\sin(100^\circ) \sec(100^\circ) \cot(100^\circ)$.

Problem 20

Determine the amplitude and period of $y = -3 \cos(\frac{\pi x}{2})$.

Problem 21

Graph $y = -3 \sec(\pi x)$. Clearly label points on the graph used in constructing the graph.

Problem 22

Find the exact value of $\csc(\cos^{-1} \frac{5}{12})$.

Problem 23

Establish the identity and SHOW ALL YOUR WORK. If I cannot tell how you got from one step to the next, you will LOSE POINTS.

$$\frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2.$$

Problem 24

Find $\sec(\tan^{-1} u)$.

Extra Credit

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\sin \theta + \sqrt{3} \cos \theta = \sqrt{3}$.